

Flexible TFT backplane development for extremely small bending radius with organic ILD and novel TFT structures

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Abstract

A new flexible low-temperature polycrystalline silicon thin film transistors (LTPS TFTs) on PI substrate with organic ILD layer, island TFT, and TFT channel perpendicular to stress direction was proposed. AMOLED display panel manufactured with organic ILD did not show any visible degradation of panel image after outward rolling cycles of 200,000 at rolling radius of 4 mm or inward folding cycles of 200,000 at folding radius of 1 mm. Another novel structure of TFT with channel perpendicular to stress direction significantly reduced change of threshold voltage of -0.03 V compared with change of threshold voltage of -0.6 V for TFT with channel parallel to stress direction for reliability test of high drain current (HDC) after 100,000 outward-bending cycles at bending radius of 3 mm. After inward-bending cycles of 200,000 at bending radius of 1 mm, reliability test results for TFT device with organic ILD layer showed that change of on-current is 10.07% for hot carrier instability (HCI) test, change of threshold voltage -0.31 V for negative bias thermal instability (NBTI) test, change of threshold voltage -0.24 V for hysteresis test, and breakdown voltage of gate insulator 7.73 MV/cm, respectively, and these performances are similar to those of TFT device with inorganic ILD (SiN_x).

KEYWORDS

flexible, island structure, organic ILD, TFT

1 | INTRODUCTION

Nowadays, active matrix organic light-emitting diodes (AMOLEDs) have received a lot of attention since they have many advantages such as self-emission, low power consumption, wide color gamut, and fast response time.^{1,2} In order to maximize these advantages of AMOLED, flexible AMOLED display can be the best candidate and highly flexible materials with low modulus needs to be developed for small bending and rolling

radius.^{3–5} Many groups have been working on the strain effect on amorphous silicon thin film transistors (TFTs) performance from the late 1990s.^{6,7} Suo et al. have reported theoretical results of mechanical strain of TFT deposited on a flexible substrate and the concept of the neutral surface with no strain applied.^{3–5} In order to achieve a device with a small bending radius, flexible materials with a small Young's modulus are required. Metals and dielectric materials commonly used in TFTs such as ITO, Mo, MoW, SiO_xN_y , SiO_x , and SiN_x have a

high modulus and are very brittle.^{7,8} Therefore, it is very difficult to achieve a small (in the range of a few mm) bending radius for an inorganic TFT without any mechanical damage to inorganic layers, which causes electrical degradation on TFTs. For example, critical bending radii of 7 mm and 9 mm were reported for SiO_xN_y and ITO, respectively, on 125 μm and 180 μm thick PET plastic substrates.^{9–11} Someya et al. also reported the bending properties on organic thin film transistors (OTFT) on 125 μm thick PEN plastic substrate.¹² Recently several companies have reported the development of foldable display which is robust up to 4 mm of bending radius.^{3,4} In this paper, mechanical and electrical properties of low-temperature polycrystalline silicon (LTPS) thin film transistors (TFTs) were evaluated. Based on the simulation result of the stress distribution in TFT backplane, several new approaches were suggested for strain-release of thin-film electron devices on flexible substrate. The new panel structure was able to be robust at a small bending radius of 1 mm, thereby enabling manufacture of foldable and rollable displays. Finally, it was suggested that organic dielectric material could tremendously reduce the stress exerted on TFT.

2 | EXPERIMENTS

P-channel low-temperature polycrystalline silicon (LTPS) thin film transistors (TFTs) on polyimide (PI) substrate with top gate structure were fabricated on 10 μm thick polyimide (PI) substrates. The OLED device is fabricated after the process of LTPS backplane. A multilayer thin film structure composed of inorganic and organic layers (called as thin film encapsulation) was used for flexible encapsulation. For bending experiments, TFT backplane on polyimide (PI) substrate was laminated on protective film (adhesive-coated polymer film) or sandwiched between two protective, which can establish the neutral plane close to TFT layer.¹³

To evaluate mechanical durability of a flexible LTPS TFT backplane, a cyclic bending test was also performed at a bending radius (R). When a TFT on PI substrate is bent, stress is distributed inside TFT backplane. Depending on the physical property of modulus for each layer, each layer experiences different stress, respectively. The position of neutral plane can be calculated according to Equation (1). Stress of plane away from neutral plane by distance of y is calculated through Equation (2). In Equations (1) and (2), F is force, σ strain, E modulus, A area, ϵ strain, y distance from neutral plane, and ρ distance from center of a circle to neutral plane in Figure 1. The simulation result showed that the biggest stress existed in interlayer dielectric

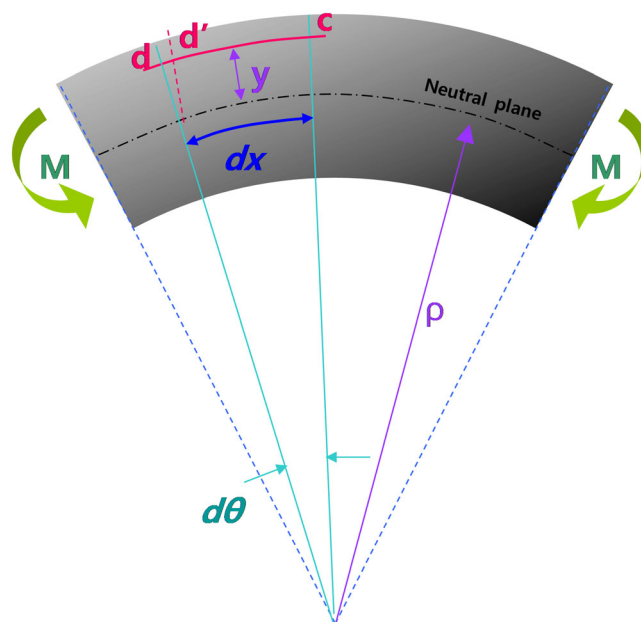


FIGURE 1 Dimension change (dd') of plane away from neutral plane by distance of y during bending.

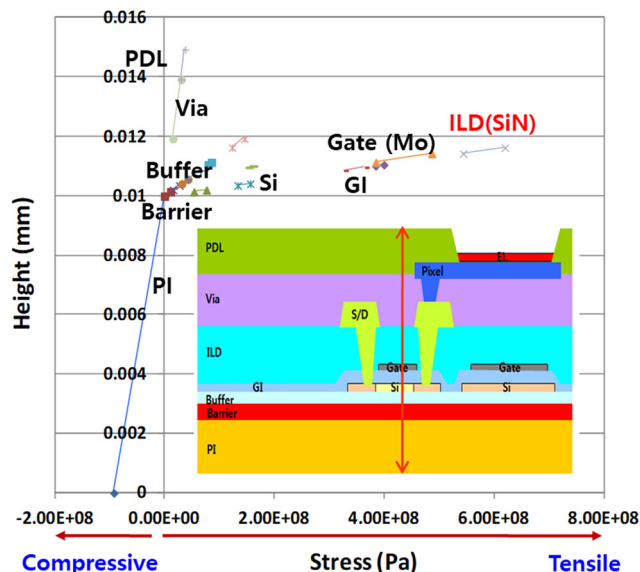


FIGURE 2 Stress distribution inside LTPS TFT device with inorganic ILD layer (SiN_x) layer.

layer (ILD) of SiN_x as shown in Figure 2, which is usually applied in conventional LTPS TFT structure. The reason why maximum stress is applied to the inorganic ILD layer of SiN_x is because the inorganic ILD of SiN_x is far from the neutral plane and its modulus is very large (300–400 GPa). This result suggested that more robust TFT structure could be achieved by replacing this brittle inorganic ILD material such as SiN_x with flexible materials of low modulus such as organic ILD materials.

$$\sum F_x = \int dF_x = \int \sigma_x dA_x = \frac{E_x}{\rho} \int y_x dA_x = 0 \quad (1)$$

$$\sum E_x y_x A_x = 0$$

$$\sigma_x = E_x \varepsilon_x = \frac{E_x}{\rho} y$$

$$dx = \rho d\theta'$$

$$dd' = cd - cd' = (\rho + y)d\theta - dx = yd\theta \quad (2)$$

$$\varepsilon_x = \frac{dd'}{dx} = y \frac{d\theta}{dx} = \frac{y}{\rho}$$

When TFT device placed in outward bending, it receives tensile stress, and when TFT device is located in inward bending, it experiences compressive stress as shown in Figure 3A. The module stack-up structure for

cyclic bending and rolling test and cyclic bending test equipment are shown in Figure 3B. New material approach to improve the flexibility of LTPS TFT back-plane is to replace inorganic inter-dielectric (ILD) layer of SiN_x with organic dielectric material with low modulus as ILD layer (phenylene or polyimide). Organic ILD layer tremendously reduced the stress on ILD layer from 5×10^7 Pa to 1.8×10^6 Pa as shown in Figure 3C and therefore significantly reduced the change of threshold voltage from 0.75 V to 0.33 V after inward-bending cycles of 400,000 at bending radius of 3 mm, and the change of capacity for organic ILD was smaller with bending cycle number in Figure 3D. As bending cycle number increases, capacity of inorganic layer decreased, which caused threshold voltage to increase according to Equation (3).¹⁴

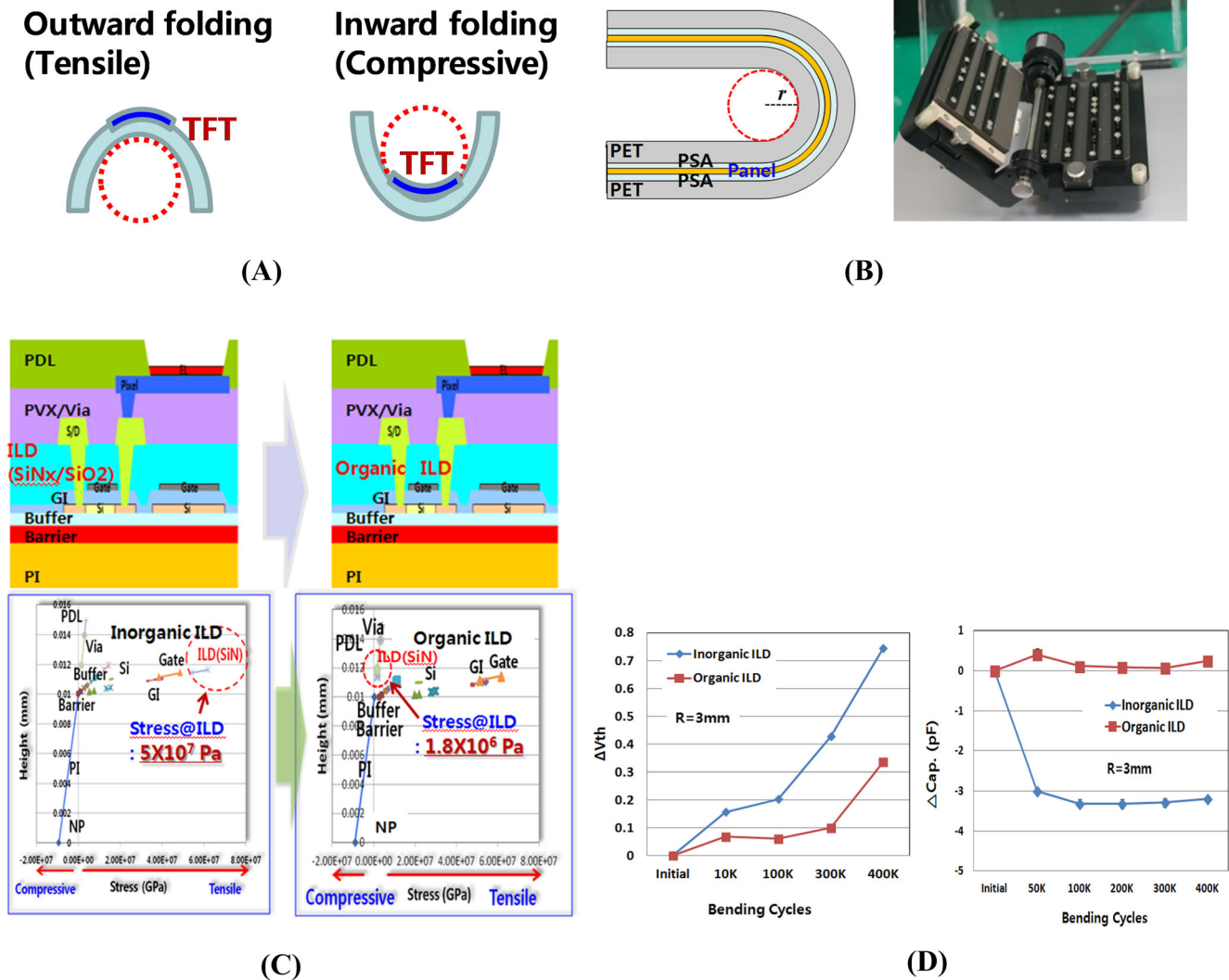


FIGURE 3 (A) Type of inward bending or outward bending, (B) module structure for cyclic bending test instrument, (C) stress distribution inside LTPS TFT with inorganic ILD (SiN_x) and organic layer (phenylene), and (D) electrical properties of LTPS TFT with inorganic ILD layer and organic ILD layer inward-bending cycles of 200,000 at bending radius of 3 mm.

$$\Delta V_{th} \propto \frac{\Delta Q_{ds}}{\Delta C_{ox}} = \frac{q \Delta N_{ds}}{C_{ox}} \quad (3)$$

Large change of capacity for inorganic ILD layer creates large change of threshold voltage as shown in Equation (3), and large change of capacity is thought to be caused by degradation of inorganic ILD layer (SiN_x) coming from mechanical bending stress.¹⁴

The electrical property of LTPS TFT for inorganic ILD layer (SiN_x) and organic ILD layer (phenylene), respectively, after inward-bending cycles of 20,000 and bending radius of 1 mm was shown in Figure 4. With organic ILD layer, change of threshold voltage was immensely lessened from 1.56 V to 0.06 V after cyclic bending. The change of mobility and subthreshold slope for organic ILD layer were $0.60 \text{ cm}^2/\text{Vs}$ and 0.001 V/decade , respectively, after cyclic bending as shown in Figure 4A, which

indicated that there was any noticeable TFT's degradation after cyclic bending.

Inorganic ILD layer, however, showed the large negative shift of threshold voltage (-0.87 V) and drop of on-current ($-5.1 \times 10^{-6} \text{ A}$) after cyclic bending of 10,000. The drop of on-current is related to the rise of defect density ($2.59 \times 10^{15}/\text{cm}^2$) (Figure 4B) in accordance to Equation (4). Q is charge, Q_f fixed charge, N_{ds} defect density, W channel width, and L channel length.

$$\begin{aligned} \Delta Q_{channel} &= \frac{\Delta I_{ds}}{\mu V_{ds}} L^2 \\ \Delta Q_f &= -\Delta Q_{channel} \\ \Delta N_{ds} &= \frac{-\Delta Q_{channel}}{qWL} = \frac{-L}{qW\mu V_{ds}} \Delta I_{ds} \end{aligned} \quad (4)$$

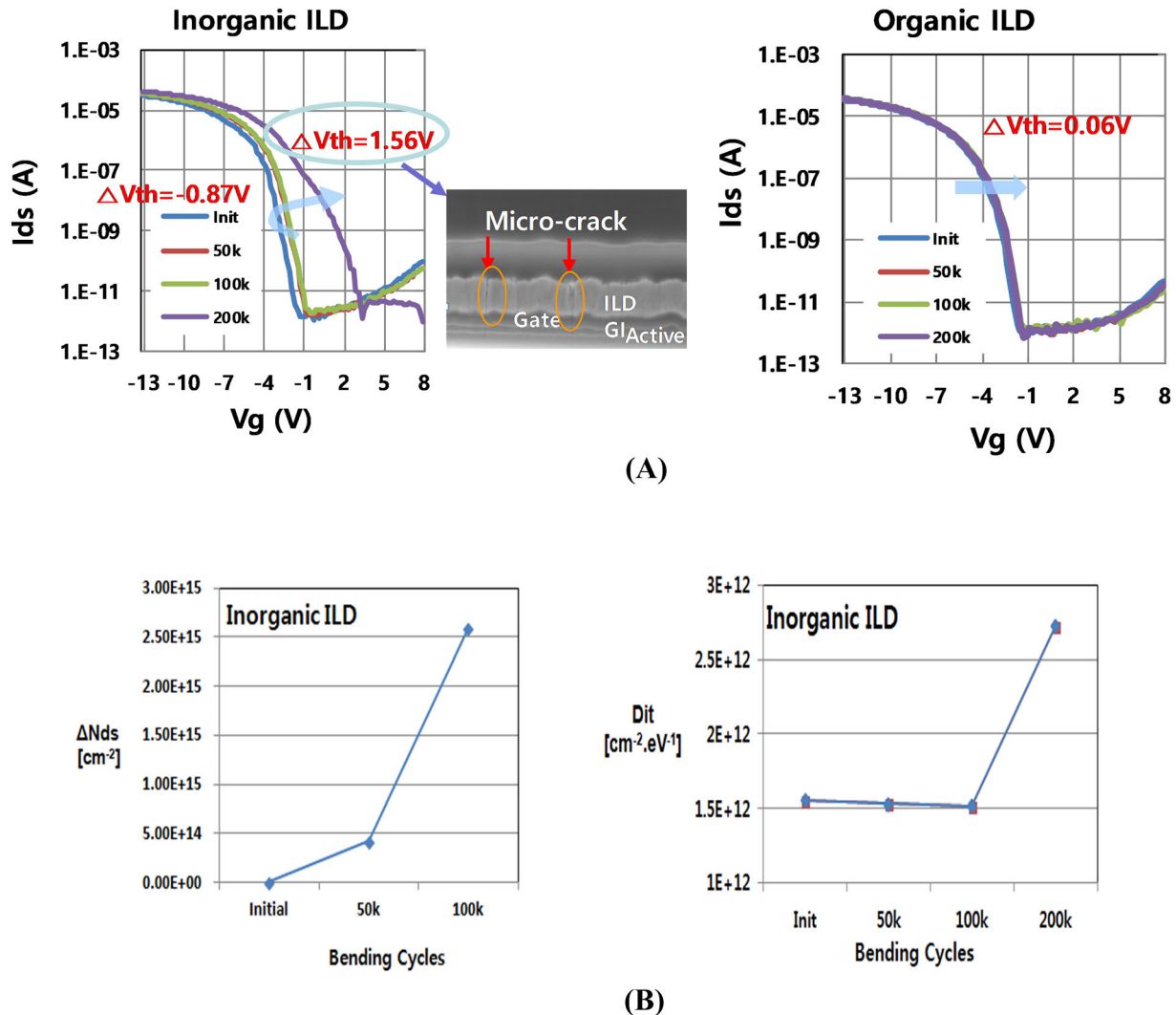


FIGURE 4 (A) Transfer curve of LTPS TFT with inorganic ILD (SiN_x) and organic ILD layer (phenylene) and (B) defect density and density of state at interfacial trap for LTPS TFT with inorganic ILD (SiN_x) after inward-bending cycles of 200,000 at bending radius of 1 mm.

After cyclic inward bending of 200,000, large positive shift of threshold voltage (1.56 V) was thought to be generated by micro-crack (permanent physical damage) of gate insulator of SiO_x and SiN_x as shown in Figure 4A, which led to leakage current and thus degradation of electrical property of TFT. Large increment of density of state at interfacial trap (D_{it}) ($2.73 \times 10^{12}/\text{cm}^2\text{eV}$) (Figure 4B) was calculated from change of subthreshold slope (SS) in accordance to Equation (5),¹⁵ where C_{ox} is capacity of oxide layer ($3.32 \times 10^{-4} \text{ F/m}^2$), q electron charge ($1.6 \times 10^{-19} \text{ C}$).

$$D_{it} = \frac{C_{ox}}{q} \left(\frac{|S|}{2.3kT} - 1 \right) \quad (5)$$

To improve the flexibility of TFT further, TFT channel direction and island TFT structure were introduced. TFT structure with active channel direction perpendicular to stress direction showed much lower change of threshold voltage ($\Delta V_{th} = -0.03 \text{ V}$) than change of threshold voltage ($\Delta V_{th} = -0.6 \text{ V}$) of TFT channel horizontal to stress direction for HDC (high drain current) stability test of $V_{gs} = -15 \text{ V}$, $V_{ds} = -20 \text{ V}$ at

25°C for 60 s after cyclic bending of 100,000 at 3 mm of bending radius because vertical TFT channel direction perpendicular to stress direction has crack propagation parallel to channel direction and therefore seems to experience by far low TFT damage compared with horizontal TFT structure with channel direction parallel to stress direction where cracks happen across channel as shown in Figure 5A. When using vertical channel TFT structure as a basic structure, island TFT structure trenching GI and ILD did not meaningfully reduce change of threshold voltage ($\Delta V_{th} = -0.045 \text{ V}$) compared with change of threshold voltage ($\Delta V_{th} = -0.084 \text{ V}$) of non-island TFT structure after cyclic bending of 200,000 at 3 mm of bending radius due to stress release effect of discontinuous structure of brittle GI and ILD layer in Figure 5B.

The photographs of 5.59-in. QHD full color top emission AMOLED display with organic ILD on a PI substrate are shown in Figure 6A. The panel with organic ILD did not have any visible defect image (line defect and point defect) after cyclic rolling of 200,000 at a rolling radius of 4 mm, also no pixel shrinkage at HTS (high temperature stability) test of 85°C , 380 h, and little luminescence

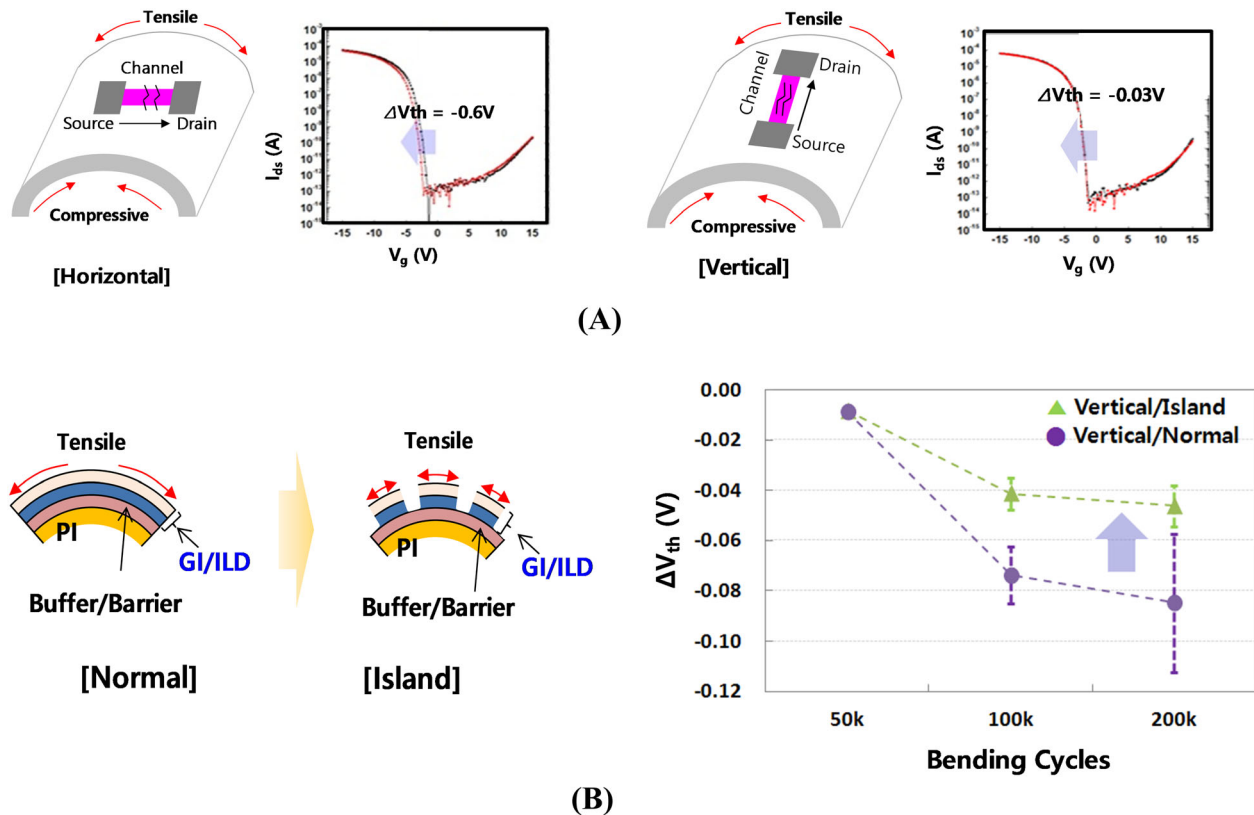


FIGURE 5 (A) Transfer curve of LTPS TFT with organic ILD layer depending on channel direction perpendicular (vertical) or parallel (horizontal) to stress direction for HDC (high drain current) reliability test after inward-bending cycles of 100,000 at bending radius of 3 mm and (B) transfer curve of LTPS TFT with island-trenched structure of inorganic GI (SiO_x) and ILD (SiN_x) after inward-bending cycles of 200,000 at bending radius of 3 mm.

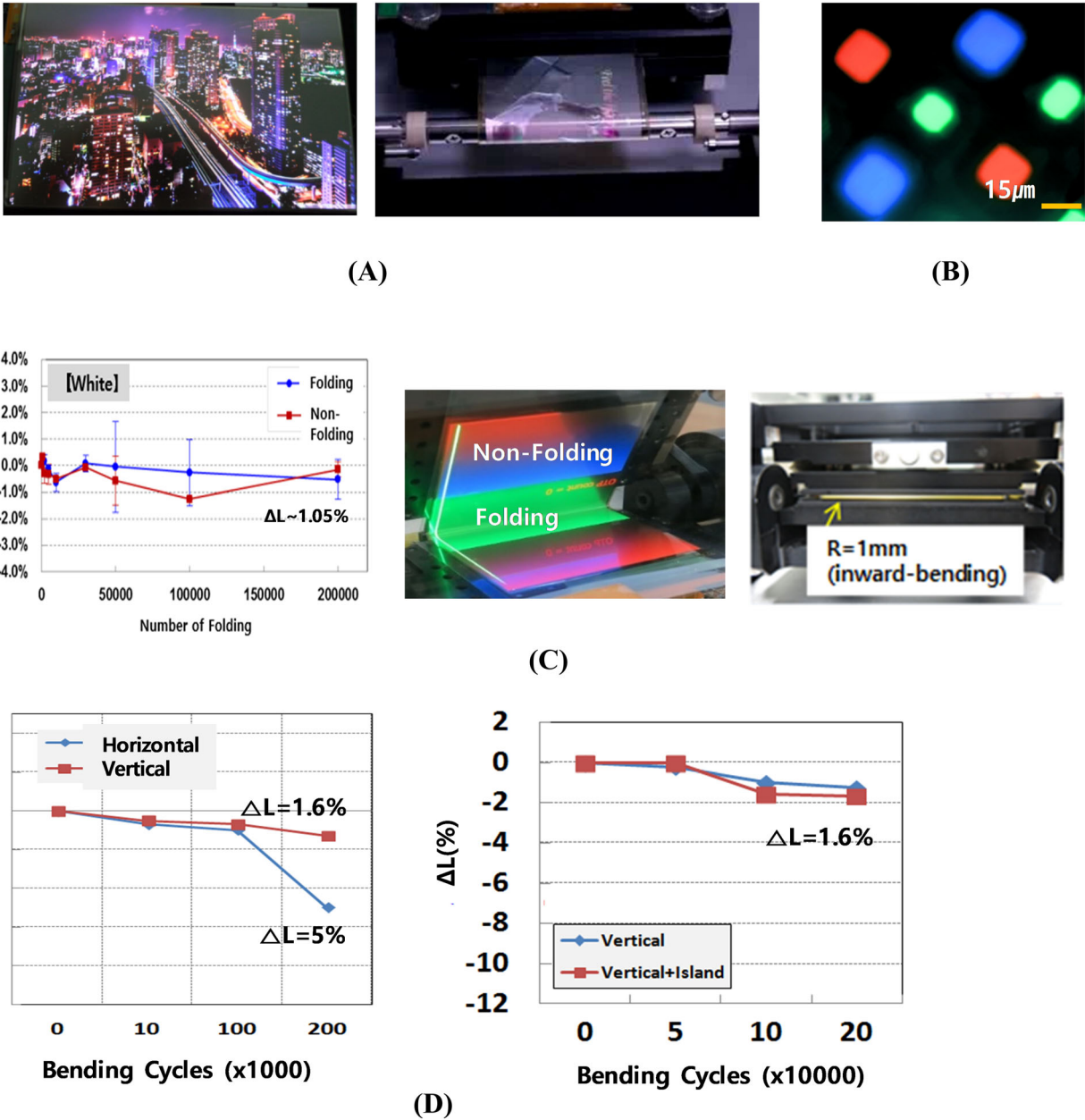


FIGURE 6 (A) Picture and rolling test image of 5.59 in. AMOLED panel with organic ILD (rolling radius of 4 mm), (B) pixel image of panel with organic ILD after HTS test of 85°C for 380 h, (C) luminescence change of panel with organic ILD after bending cycles of 200,000 at bending radius of 1 mm, and (D) luminescence change of panel based on inorganic ILD (SiN_x) with different TFT structures of vertical channel, horizontal channel, and island patterning of inorganic GI (SiO_x) and ILD (SiN_x) after bending cycles of 200,000 at bending radius of 3 mm.

TABLE 1 Electrical stability test results with organic ILD layer after inward-bending cycles of 200,000 at bending radius of 1 mm.

	HCI	NBTI	Hysteresis	GIBV
Results	$10.07 \pm 1.8\%$	$-0.31 \pm 0.21 \text{ V}$	$-0.24 \pm 0.05 \text{ V}$	$7.73 \pm 0.47 \text{ MV/cm}$

change ($\Delta L \sim 1.05\%$) after cyclic bending (folding) of 200,000 at bending radius of 1 mm as presented in Figure 6B,C.

The 4.65-in. panel with vertical TFT channel and island TFT structure indicated very small change of luminescence ($\Delta L \sim 1.6\%$), namely, significantly increase the stress release, compared to change of luminescence ($\Delta L \sim 5\%$) of panel with conventional TFT structure after cyclic bending of 200,000 at bending radius of 3 mm as indicated in Figure 6D. Effect of active channel direction seems to be greater than effect of island TFT structure regarding stress release effect because the change of luminescence ($\Delta L \sim 1.6\%$) for vertical TFT channel structure plus island TFT structure is similar to that of vertical TFT structure. This result of change of luminescence is almost same as TFT test result of change of threshold voltage.

Electrical stability tests such as HCI (hot carrier instability) of $V_{gs} = -12.2$ V, $V_{ds} = -22.4$ V at 25°C for 4000 s, NBTI (negative bias temperature instability) of $V_g = -14$ V at temperature of 70°C for 10 h, hysteresis of $V_g = 8$ V $\leftrightarrow$ -13 V, and GIBV (gate insulator breakdown voltage) after cyclic mechanical stress were previously performed in order to estimate the robustness of TFT with organic ILD layer against residual stress after 200,000 inward-bending cycles at bending radius of 1 mm as shown in Table 1. All electrical stability test results of HCI, NBTI, hysteresis, and GIBV were with our specifications, and these results indicated that TFT with organic ILD layer is very robust even at bending cycles of 200,000 at bending radius of 1 mm.

3 | CONCLUSION

Several remarkable new improved flexible backplane structures with flexible organic ILD layer, vertical channel TFT structure, and island TFT structure approach have been proposed to effectively reduce mechanical stress in LTPS TFT under bending and rolling deformation. With this new structure, TFT backplane and display panel are able to be bent up to an ultimate bending radius of without any TFT degradation, panel characteristics degradation, and any mechanical damage 1 mm after inward-bending cycles of 200,000. The new strain-release structure enables the manufacture of electronic devices demanding ultimate mechanical flexibility and robustness. By combining these several novel stress-release structures, we are able to manufacture extremely flexible AMOLED display (including foldable and rollable display) based on LTPS TFT backplane even at small bending radius of 1 mm and rolling radius of 4 mm.

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